

Cumulative Distributions

Worked Z-Score Problem

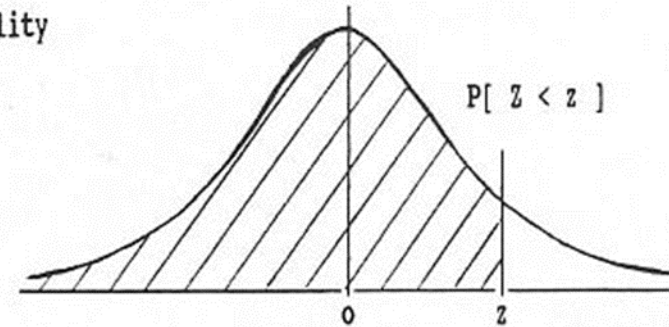
Cumulative Proportions

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability
up to the standardised normal value z
i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}z^2) dz$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5518	0.5557	0.5597	0.5637	0.5677	0.5717	0.5757
0.2	0.5797	0.5837	0.5877	0.5917	0.5957	0.5997	0.6037	0.6077	0.6117	0.6157
0.3	0.6197	0.6237	0.6277	0.6317	0.6357	0.6397	0.6437	0.6477	0.6517	0.6557
0.4	0.6597	0.6637	0.6677	0.6717	0.6757	0.6797	0.6837	0.6877	0.6917	0.6957
0.5	0.6997	0.7037	0.7077	0.7117	0.7157	0.7197	0.7237	0.7277	0.7317	0.7357
0.6	0.7397	0.7437	0.7477	0.7517	0.7557	0.7597	0.7637	0.7677	0.7717	0.7757
0.7	0.7797	0.7837	0.7877	0.7917	0.7957	0.7997	0.8037	0.8077	0.8117	0.8157
0.8	0.8197	0.8237	0.8277	0.8317	0.8357	0.8397	0.8437	0.8477	0.8517	0.8557
0.9	0.8597	0.8637	0.8677	0.8717	0.8757	0.8797	0.8837	0.8877	0.8917	0.8957
1.0	0.8997	0.9037	0.9077	0.9117	0.9157	0.9197	0.9237	0.9277	0.9317	0.9357
1.1	0.9397	0.9437	0.9477	0.9517	0.9557	0.9597	0.9637	0.9677	0.9717	0.9757
1.2	0.9797	0.9837	0.9877	0.9917	0.9957	0.9997	1.0000	1.0000	1.0000	1.0000

The cumulative proportion for a value x in a distribution is the proportion of observations in the distribution that are less than or equal to Z .

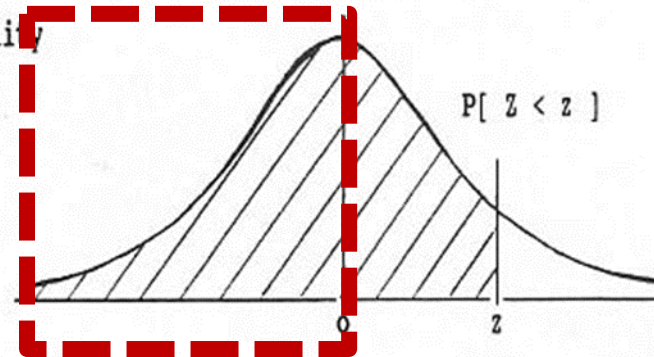
Worked Z-Score Problem

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability
up to the standardised normal value z
i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015

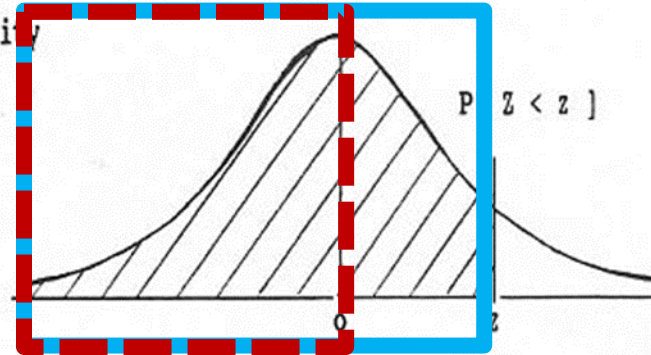
Worked Z-Score Problem

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability
up to the standardised normal value z
i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	.8413	.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015

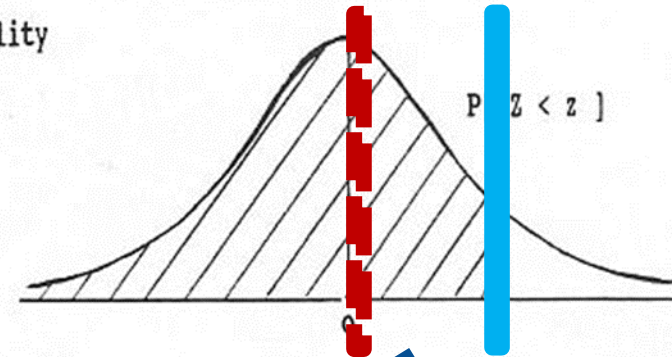
Worked Z-Score Problem

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability
up to the standardised normal value z
i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}z^2) dz$$



z 0.00 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08 0.09

$$.8413 - .500 = .3413$$

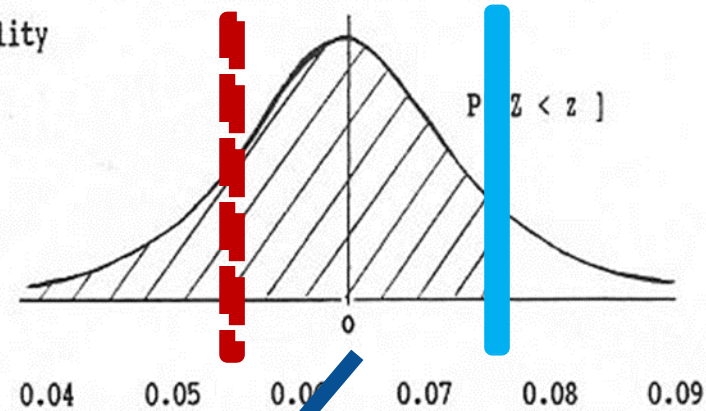
Worked Z-Score Problem

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability
up to the standardised normal value z
i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}z^2) dz$$



$$.8413 - .500 = .3413 * 2 = .68$$

Worked Z-Score Problem

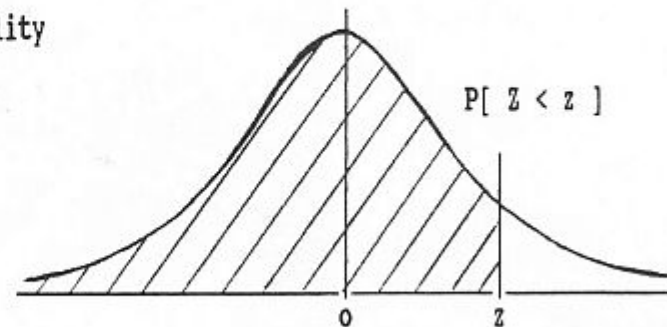
- B) What would these proportions be in an exactly Normal distribution?
- $\pm$ TWO SD

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability
up to the standardised normal value z
i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}z^2) dz$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-----	------	------	------	------	------	------	------	------	------	------

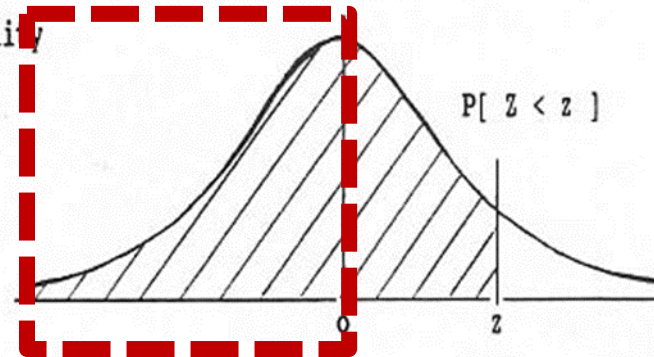
Worked Z-Score Problem

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability
up to the standardised normal value z
i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015

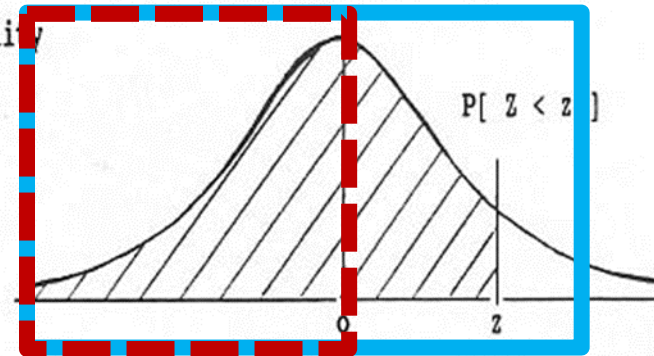
Worked Z-Score Problem

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability
up to the standardised normal value z
i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}z^2) dz$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857

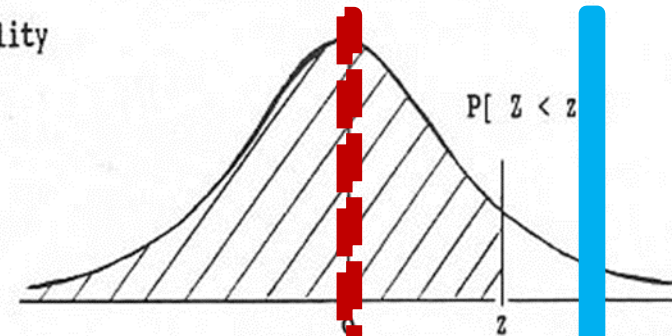
Worked Z-Score Problem

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability
up to the standardised normal value z
i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz$$



z 0.00 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08 0.09

$$.9773 - .500 = .4773$$

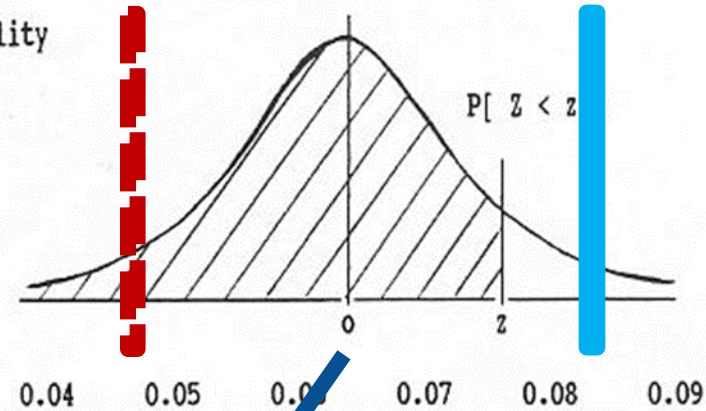
Worked Z-Score Problem

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability
up to the standardised normal value z
i.e.

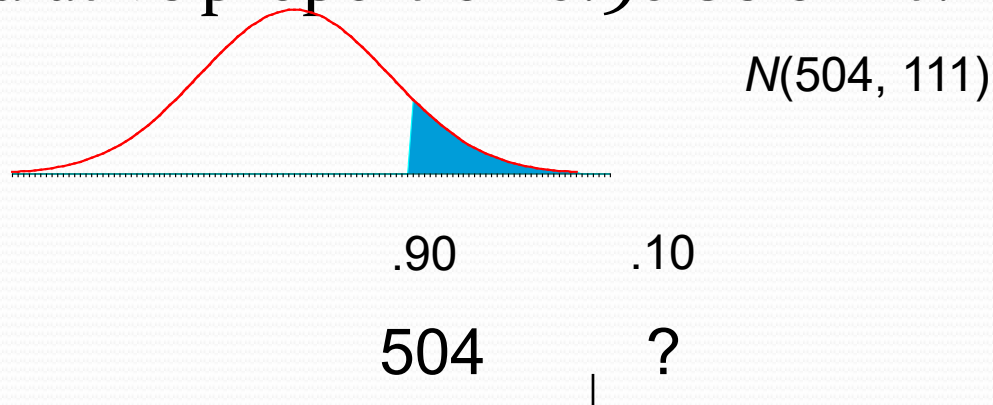
$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz$$



$$.9773 - .500 = .4773 * 2 = .9546$$

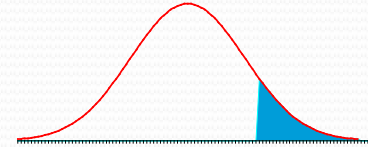
Starting from a proportion

- SAT reading scores for a recent year are distributed according to an $N(504, 111)$ distribution.
- How high must a student score in order to be in the top 10% of the distribution?
- In order to use table A, equivalently, what score has cumulative proportion 0.90 below it?



Normal Calculations (2 of 3)

How high must a student score in order to be in the top 10% of the distribution?



- Look up the closest probability (closest to 0.10) in the table.
- Find the corresponding **standardized score**.
- The value you seek is *that many standard deviations from the mean*.

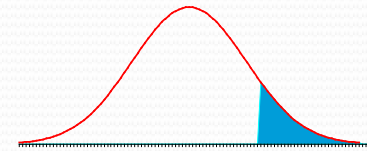
.10
|
504 ?

z	.07	.08	.09
1.1	.8790	.8830	.8830
1.2	.8997	.9015	.9015
1.3	.9147	.9162	.9177

$$z = 1.28$$

Normal Calculations (3 of 3)

How high must a student score in order to be in the top 10% of the distribution?



$$z = 1.28$$

504 .10
| ?

We need to “unstandardize” the z-score to find the observed value (x):

$$z = \frac{x - \mu}{\sigma}$$



$$x = \mu + z\sigma$$

$$\begin{aligned} x &= 504 + z \cdot 111 \\ &= 504 + [(1.28) \cdot 111] \\ &= 504 + (142.08) = 646.08 \end{aligned}$$

A student would have to score at least 646.08 to be in the top 10% of the distribution of SAT reading scores for this particular year.

“Backward” Normal Calculations

USING TABLE A GIVEN A NORMAL PROPORTION

STEP 1: State the problem in terms of the given proportion. **Draw a picture** that shows the Normal value, x , that you want in relation to the cumulative proportion.

STEP 2: Use Table, the fact that the total area under the curve is 1, and the given area under the standard Normal curve to find the corresponding z -value.

STEP 3: Unstandardize z to solve the problem in terms of a non-standard Normal variable x .

Functions in R

`dnorm()` gives the density,

`pnorm()` gives the cumulative density function,

- computes the probability that a normally distributed random number will be less than that number

`qnorm()` gives the quantile function, and

- (is the inverse of `pnorm`, give it a probability,
- it produces the number whose cumulative distribution matches the probability

`rnorm()` generates random deviates.

Illustrations with R

- Throwback: Sampling and Standard Error
- Robustness